An *ab-initio* Diagrammatic Monte Carlo prototype for the LiF electron polaron

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Abstract

This note fixes the **goals**, deliberate **non-goals**, and **deferred** extensions of the `rmc-ab-initio` project: a diagrammatic Monte Carlo (DiagMC) prototype, built on the `ironmc` engine, that replaces the analytic Fröhlich electron–phonon vertex $|g(\mathbf{q})|^2 \propto \alpha/q^2$ with the *real* electron–phonon matrix elements $g(\mathbf{k}, \mathbf{q}, \nu)$ obtained from density-functional perturbation theory and Wannier interpolation (Quantum ESPRESSO + EPW). The first validation material is the electron polaron in LiF, a wide-gap ionic insulator that sits close to the textbook Fröhlich limit and therefore provides a built-in analytic cross-check. The scope below is the *first* prototype; Section 4 records what is intentionally postponed.

1 Overview

`ironmc` is a Rust workspace of DiagMC crates that sample the imaginary-time Green's function of a polaron. This project adds `rmc-diagmc`, a state-generic DiagMC engine exposing a `Model` trait (electron dispersion, phonon dispersion, complex e–ph vertex, phonon-momentum proposal), plus a standalone *ab-initio* consumer `rmc-ab-initio`. Two models implement the trait: the closed-form *analytic* Fröhlich model, and the *ab-initio* model whose material-specific electron–phonon (e–ph) vertex $g(\mathbf{k}, \mathbf{q}, \nu)$ is read from a real DFT/DFPT + Wannier calculation. The ground-state energy is read off the large-τ decay of $G(\tau)$; there is no self-consistent iteration.

The data pipeline is

$$\underbrace{\text{pw.x, ph.x}}_{\text{Quantum ESPRESSO}} \longrightarrow \underbrace{\text{epw.x}}_{\text{Wannier e–ph tensors}} \longrightarrow \underbrace{\text{Rust reader + reconstruction}}_{\texttt{crates/apps/rmc-ab-initio}} \longrightarrow \text{DiagMC.}$$

DFPT supplies the phonons, the Born effective charges $\mathbf{Z}^*_\kappa$, and the electronic dielectric tensor $\boldsymbol{\varepsilon}_\infty$; EPW supplies the e–ph coupling in the localized Wannier representation $g(\mathbf{R}_e, \mathbf{R}_p)$, from which $g(\mathbf{k}, \mathbf{q}, \nu)$ is recovered by an inverse Fourier transform.

Why LiF. Rocksalt LiF has a single, non-degenerate conduction-band minimum at Γ, one dominant longitudinal-optical (LO) branch, and an intermediate-to-strong Fröhlich coupling $\alpha \approx 4.8$. It is therefore the cleanest first material: nearly the textbook Fröhlich case, with a genuine analytic limit to validate against.

Units. The ab-initio model works in *physical* units — energies in eV, imaginary time in eV^{-1}. The electron dispersion $\varepsilon(\mathbf{k})$ is the real Wannier-interpolated conduction band from $H(\mathbf{R})$ (not a parabola), and the phonons are the real branches $\omega_{\mathbf{q}\nu}$ from the polar-restored dynamical matrix,

with *all* modes retained. The Fröhlich coupling is still computed from the macroscopic constants as a cross-check,

$$\alpha \;=\; \frac{e^2}{\hbar}\sqrt{\frac{m^*}{2\hbar\omega_{\mathrm{LO}}}}\left(\frac{1}{\varepsilon_\infty}-\frac{1}{\varepsilon_0}\right) \;\approx\; 4.76 \quad \text{(LiF)}, \tag{1}$$

which fixes the weak-coupling target $E^{(2)} \approx -\alpha\,\hbar\omega_{\mathrm{LO}}$ against which the vertex scale is checked (Section 9, P8).

2 Goals

G1 — Real vertex. Drive the DiagMC sampler with the *ab-initio* e–ph vertex in place of the analytic one. The vertex is decomposed as

$$g_{mn\nu}(\mathbf{k},\mathbf{q}) \;=\; g^{\mathcal{S}}_{mn\nu}(\mathbf{k},\mathbf{q}) \;+\; g^{\mathcal{L}}_{mn\nu}(\mathbf{k},\mathbf{q}), \tag{2}$$

the short-range (Wannier-interpolated) part plus the long-range polar part.

G2 — Generic engine, native model. Factor the sampler into a state-generic `rmc-diagmc` engine (`Runner`, `Update`, `MetropolisKernel`, decay estimator) behind a `Model` trait. The ab-initio model is a *native* implementation of that trait — real $\varepsilon(\mathbf{k})$, $\omega_{\mathbf{q}\nu}$, complex $g(\mathbf{k},\mathbf{q},\nu)$, and a $|g|^2$-weighted phonon-momentum proposal — *not* a reweighting of the Fröhlich envelope. The Fröhlich model is a second implementation of the same trait, which validates the engine at stage 0 (Section 5).

G3 — Correct polar long range. Restore the long-range Fröhlich $1/q$ divergence that dominates the LiF polaron. EPW was run with `lpolar=.true.`, which *subtracts* the long-range part before Wannierization — from *both* the e-ph vertex and the dynamical matrix — so $g^{\mathcal{S}}$ from the tensor is short-ranged and both the vertex term $g^{\mathcal{L}}$ (Eq. (3)) and the dynamical-matrix term $D^{\mathcal{L}}$ (Eq. (4)) must be added back analytically. The latter is what restores the physical TO frequency and the LO–TO splitting; without it the phonon zero-point factor $(2M_\kappa\omega)^{-1/2}$ in $g^{\mathcal{L}}$ is mis-scaled (Section 9, P7). *Done:* both implemented as from-scratch ports of EPW's `rgd_blk_epw` and `rgd_blk` (Ewald parameters $\Lambda_{\mathrm{Ew}} = 1$, $g_{\max} = 14$, $e^2 = 2$ Ry) and validated to machine precision against the coarse Bloch `.epb` and dynamical matrix `dynq` (Section 9, P6–P7).

G4 — Clean Rust reader. Implement the EPW-file readers and the Wannier reconstruction from scratch in Rust, with no Python at runtime. The existing Python post-processor (`patt126-ab-initio-diagmc`) is used as a *reference only*, not copied; the rewrite also fixes a latent index assumption in the reference.

G5 — Two-front validation. Benchmark the LiF electron polaron against *both* (i) Fröhlich-limit DiagMC energetics at α_{LiF} (Mishchenko *et al.*, 2000), and (ii) *ab-initio* LiF polaron numbers (Sio, Verdi, Poncé, Giustino, 2019).

The long-range term used for G3 (Verdi–Giustino form) is

$$g^{\mathcal{L}}_{mn\nu}(\mathbf{k},\mathbf{q}) \;=\; i\,\frac{4\pi e^2}{\Omega}\sum_{\kappa}\left(\frac{\hbar}{2M_\kappa\omega_{\nu\mathbf{q}}}\right)^{1/2}\sum_{\mathbf{G}\neq-\mathbf{q}}\frac{(\mathbf{q}+\mathbf{G})\cdot\mathbf{Z}^*_\kappa\cdot\mathbf{e}_{\kappa\nu}(\mathbf{q})}{(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\varepsilon}_\infty\cdot(\mathbf{q}+\mathbf{G})}\,e^{-i(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\tau}_\kappa}\,e^{-\frac{(\mathbf{q}+\mathbf{G})\cdot\boldsymbol{\varepsilon}_\infty\cdot(\mathbf{q}+\mathbf{G})}{4\Lambda}}\left[U_{\mathbf{k}+\mathbf{q}}U^\dagger_{\mathbf{k}}\right]_{mn}, \tag{3}$$

with Ω the cell volume, $M_\kappa, \boldsymbol{\tau}_\kappa$ the atomic masses and positions, $\mathbf{e}_{\kappa\nu}$ the phonon eigenvectors, and Λ an Ewald range-separation parameter. The final factor $[U_{\mathbf{k}+\mathbf{q}} U_{\mathbf{k}}^\dagger]_{mn}$ is the electron-band rotation into the smooth (Wannier) gauge; at the Γ-slice ($\mathbf{k} = \Gamma$, single conduction band) it reduces to the conduction-band Bloch overlap $\langle u_{\Gamma+\mathbf{q}} | u_\Gamma \rangle$. Restoring it is essential to the exact short/long decomposition of Eq. (2); see Section 9, P6. As $q \to 0$ the term behaves as $1/q$, reproducing the Fröhlich tail $|g_F(\mathbf{q})|^2 \propto \alpha/q^2$.

The *same* dipole physics enters the dynamical matrix: EPW likewise removes the rigid-ion (dipole–dipole) long-range from D before Wannierization, so the interpolated phonons are short-ranged — the optical triplet stays degenerate at Γ at an unphysical frequency with no LO–TO splitting. The term to add back (Gonze form, EPW `rgd_blk`) is, with $\mathbf{Q} \equiv \mathbf{q} + \mathbf{G}$,

$$D^{\mathcal{L}}_{\kappa\alpha,\kappa'\beta}(\mathbf{q}) \;=\; \frac{4\pi e^2}{\Omega} \sum_{\mathbf{G}} \frac{e^{-\mathbf{Q}\cdot\boldsymbol{\varepsilon}_\infty\cdot\mathbf{Q}/4\Lambda}}{\mathbf{Q}\cdot\boldsymbol{\varepsilon}_\infty\cdot\mathbf{Q}} \,(\mathbf{Q}\cdot\mathbf{Z}^*_\kappa)_\alpha \,(\mathbf{Q}\cdot\mathbf{Z}^*_{\kappa'})_\beta \, e^{i\mathbf{Q}\cdot(\boldsymbol{\tau}_\kappa - \boldsymbol{\tau}_{\kappa'})} \;-\; \delta_{\kappa\kappa'}\, S_{\kappa\alpha\beta}, \tag{4}$$

where the self-term $S_{\kappa\alpha\beta} = \sum_{\kappa''} D^{\mathcal{L}}_{\kappa\alpha,\kappa''\beta}(\mathbf{0})$ (the same sum at $\mathbf{q} = 0$, summed over the second atom) is subtracted on the diagonal blocks to enforce the acoustic sum rule. Diagonalizing $D^{\mathcal{S}} + D^{\mathcal{L}}$ (mass-reduced) gives the physical branches $\omega_{\mathbf{q}\nu}$ and eigenvectors $\mathbf{e}_{\kappa\nu}(\mathbf{q})$ that enter Eq. (3).

3 Non-goals (deliberate scope of the first prototype)

Several early non-goals of the *first* prototype have since been **lifted**: the Γ-slice vertex, the single Einstein branch, the "no tensor compression" rule, the reweighted sampler, the Fröhlich-unit engine constants, and the "no GPU path" rule (N7). The model now uses the full electron-momentum vertex $g(\mathbf{k}, \mathbf{q}, \nu)$, all phonon branches, an SVD-compressed Wannier tensor evaluated on a grid, a native $|g|^2$ momentum proposal, physical units, and a validated CUDA backend (Section 8). What remains deliberately out of scope:

N1 — Electron polaron only. The single, non-degenerate conduction-band minimum at Γ. The hole/valence polaron (degenerate top valence band) is out of scope.

N8 — Single material. LiF is the sole validation target; this is not yet a general multi-material framework.

(The retired non-goals keep their original labels N2–N7 where referenced elsewhere; they are now realized features, recorded as done in Section 4.)

4 Deferred to future work

Done. The full-$\mathbf{k}$ vertex with SVD compression and on-grid evaluation (former N2/N4), all dispersive branches $\omega_\nu(\mathbf{q})$ (N3), the non-parabolic Wannier dispersion $\varepsilon(\mathbf{k})$ via the generic `Model` trait (N6), the native $|g|^2$ momentum sampler (N5), and the CUDA backend (former N7/D3, Section 8) are now *implemented*.

Remaining.

D1 — Grid convergence. *Done* for the energy. The vertex is tabulated on a uniform n^3 grid; the discrete BZ sum underweights the near-Γ $1/q^2$ polar region, so the polaron energy converges from above with a leading $1/n$ error (at n=8, $E_0 \approx -0.122$ eV). Ladders $n = 20 \ldots 120$ at

$\tau = 60$ and $\tau = 240$ eV^{-1} and their extrapolation are in Section 6. Still open: the τ=480 guard at 120^3, and a Γ-cell (Fröhlich control-variate) correction that would remove the leading $1/n$ term rather than fit it.

D2 — Hole polaron. The degenerate valence-band-maximum polaron (lifting N1), needing a multi-band, degenerate (matrix-product sign-fix) treatment.

D4 — More materials. Generalize beyond LiF once the LiF prototype is validated (lifting N8).

5 Validation stages

The goals of Section 2 are checked in stages of increasing physical content:

1. **Engine (stage 0).** The analytic Fröhlich model, run through `rmc-diagmc` and the decay estimator, reproduces Mishchenko *et al.* (2000): $E_0(\alpha{=}1) = -1.013$ and $E_0(\alpha{=}2) = -2.08$ within 3σ, sign $\equiv 1$. This validates the sampler, estimator, and sign tracking independently of the ab-initio vertex.

2. **Vertex scale.** The ab-initio second-order energy $E^{(2)}(\Gamma)$ recovers the Fröhlich weak-coupling target $-\alpha\,\hbar\omega_{\mathrm{LO}}$ (up to the coarse-grid underestimate), confirming the $(2M_\kappa\omega)^{-1/2}$ vertex normalization (Section 9, P8).

3. **Compression convergence.** The all-orders E_0 is certified against the compression *in the sampler itself*: the production rank budget must agree with the uncompressed vertex to within a stated tolerance, $\Delta_{\mathrm{rank}} = E_0(\mathrm{budget}) - E_0(\mathrm{uncompressed})$, and the measured Δ_{rank} is quoted as a systematic on every production number (Section 6).

4. **Material comparison.** The grid-ladder formation energy against FEP-DMC (Luo, Park, Bernardi, 2025): their $E_0(20^3) = -0.2479$ eV (from their example output, a single run at their effective $\tau \approx 232$ eV^{-1}) and -0.408 eV in the $1/n \to 0$ limit (linear fit through their three largest grids). At matched $\tau = 240$ eV^{-1} we measure $E_0(20^3) = -0.2506 \pm 0.0003$ eV, 2.7 meV deeper; the extrapolated $E_\infty = -0.4063 \pm 0.0006\,(\mathrm{stat}) \pm 0.0008\,(\mathrm{fit\ model})$ eV is 1.7 meV shallower (Section 6). The two ladders therefore differ in slope by a few meV over $n = 20 \to \infty$, plausibly from dataset provenance (pseudopotentials, cutoffs, their rank budget); agreement is at the 1% level, and a meV-level claim is not supported. Neither number includes the remote-band (Allen–Heine–Cardona) correction, which FEP-DMC adds separately to reach -0.538 eV.

6 Discretization, compression, and the thermodynamic limit

The object the sampler actually solves is not the continuum polaron Hamiltonian but its restriction $\hat{H}_n$ to a uniform n^3 Brillouin-zone grid shared by electrons and phonons. The grid is closed under momentum addition, so every diagram's momentum bookkeeping stays on it; the per-arc momentum measure is the lattice normalization $\frac{1}{N_q}\sum_{\mathbf{q}\neq\Gamma}$ with $N_q = n^3$ (matching the standard first-principles self-energy convention), and the DiagMC solves each $\hat{H}_n$ *numerically exactly*. All physical statements are then statements about the sequence $E_0(n)$: because the discrete sum underweights the $1/q^2$ polar region within a $1/n$-neighbourhood of Γ, $E_0(n)$ converges from above with a leading error linear in $1/n$, and the thermodynamic limit is read off a $1/n \to 0$ extrapolation over a ladder of grids — the same methodology as FEP-DMC, whose LiF benchmarks are $E_0(20^3) = -0.248$ eV and -0.408 eV extrapolated.

The LiF ladder and its error budget. Banked rungs: $\tau = 60\,\mathrm{eV}^{-1}$ at $n = 20, 32, 40, 50, 60, 80, 100, 120$ and $\tau = 240\,\mathrm{eV}^{-1}$ at $n = 20, 40, 60, 80, 100, 120$ (runbook `notebooks/LiF-e0-gamma-extrapolation.ipynb`). Only the leading $u = 1/n$ term has a physical origin (the missing Γ cell), so the model $E_0(u) = E_\infty + Cu + Du^2$ is fitted by weighted least squares, and the spread over fit models and windows is quoted as a separate systematic. For $\tau = 240$ the quadratic fit over all rungs gives $E_\infty = -0.4063 \pm 0.0006$ eV ($\chi^2_{\mathrm{red}} = 1.4$). Acceptable alternatives ($\chi^2_{\mathrm{red}} \le 2.2$, own error ≤ 2 meV: linear from $n \ge 40$, linear through the three largest grids as in FEP-DMC, quadratic from $n \ge 40$) span $-0.4064 \ldots -0.4048$ eV, i.e. a fit-model systematic of ± 0.8 meV. Underdetermined fits (cubic on five points, own error ≈ 6 meV) are uninformative rather than competing models, and are excluded. An independent route agrees: the τ shift $E_0(\tau 240) - E_0(\tau 60)$ is n-independent for $n \ge 40$ (-2.83 ± 0.13 meV), and adding it to the better-conditioned eight-rung $\tau 60$ quadratic fit (-0.4030 ± 0.0002 eV) gives -0.4058 eV. The compression systematic $\Delta_{\mathrm{rank}}(800) = +0.5 \pm 0.3$ meV (at 20^3) is below the fit-model systematic. The τ systematic is certified by a $\tau = 480\ \mathrm{eV}^{-1}$ guard at $n = 120$ (pending). Current quote: $E_\infty = -0.4063 \pm 0.0010$ eV (stat. and fit model combined), before the τ guard.

Compression as a controlled, quoted approximation. The short-range vertex is compressed to a global rank budget (the largest singular values across all band pairs); the long-range polar term is always exact. The acceptance criterion is *never* element-wise reconstruction error — it is convergence of the physical observable, certified in the sampler itself:

$$\Delta_{\mathrm{rank}} \;=\; E_0(\mathrm{budget}) - E_0(\mathrm{uncompressed}),$$

measured by two otherwise-identical DMC runs and quoted as a systematic error alongside the statistical one on every production result. Two empirical lessons fixed this protocol. First, the cheap proxy — the second-order energy $E^{(2)}$ at $\mathbf{k} = \Gamma$ — systematically *overestimates* the true sampler systematic by a factor of 3–10, because the sampler averages the compression error over the whole zone while the proxy probes a single $\mathbf{k}$-row; the proxy is useful for scanning, but only the in-sampler measurement decides. Second, convergence with the global budget is *non-monotonic* (singular values rank global tensor weight, not importance for a given observable), so "two nearby budgets agree" is not evidence of convergence — only the comparison against the uncompressed vertex is. For the current LiF dataset the certified production budget is 800 of 1600 components with $\Delta_{\mathrm{rank}} = 1.0 \pm 0.8$ meV; the FEP-DMC-sized budget (≈ 300) measured a genuine 4.6 ± 0.8 meV and was rejected.

Dataset provenance. The DFPT/Wannier inputs use $8 \times 8 \times 8$ coarse k- and q-grids, matching FEP-DMC's LiF setup. Upgrading from the initial $6 \times 6 \times 6$ deepened $E_0(8^3)$ by ~ 11 meV — a pure interpolation-quality correction, worth recording because it is the same order as the compression and statistical errors being controlled above. One physical subtlety appeared with the finer grid: it contains the W star, where the two lowest conduction bands are exactly degenerate, making per-band comparisons gauge-ambiguous there (the round-trip validation skips those points; the sampler itself is unaffected, since it consumes gauge-invariant $|g|^2$ combinations).

7 Performance: the ab-initio vertex cache

The full electron-momentum vertex — the former Γ-slice non-goal (N2), now lifted — cannot be reduced to the scalar $|g(\Gamma, \mathbf{q}, \mathrm{LO})|^2$. Instead the Wannier e–ph tensor is compressed by a per-band-pair truncated SVD into rank-r factors $u_g(\mathbf{k})$, $v_g(\mathbf{q}, \nu)$, tabulated once on the q-grid; a vertex

evaluation is then the rank-r contraction

$$g(\mathbf{k}, \mathbf{q}, \nu) \;=\; \sum_{g=1}^{r} s_g \, u_g(\mathbf{k}) \, v_g(\mathbf{q}, \nu) \;+\; g^{\mathcal{L}}(\mathbf{k}, \mathbf{q}, \nu),$$

called millions of times per DiagMC chain, so its cost sets the wall-clock. On an 8^3 grid at $r \sim 150$ the factor tables are ≈ 0.18 GiB — far larger than cache — so the memory layout of the component axis g, not the arithmetic, dominates.

Cache-hostile layout (before). Storing the factors component-major, $u[g][\mathbf{k}]$ and $v[g][\mathbf{q}][\nu]$, places successive components of a single vertex sum ~ 49 KB apart, so each of the r terms is a separate cache miss and the inner contraction is bound entirely by memory latency.

Fix: transpose the component axis inward. Storing the factors $\mathbf{k}$-major / $(\mathbf{q}, \nu)$-major — $u[\mathbf{k}][g]$, $v[\mathbf{q}][\nu][g]$ — makes the r components of one vertex evaluation contiguous, so the contraction streams two cache-line runs instead of striding r scattered lines. The precompute is reorganized to parallelize over $\mathbf{k}$ (rather than over g) so it writes this layout directly.

Effect. For the n_{sv} convergence sweep (500k steps $\times$ 10 chains, 8^3 grid), a single rank drops from 258 $\rightarrow$ 23 s at r=50 and 1033 $\rightarrow$ 54 s at r=100 — an 11–19$\times$ wall-clock reduction that grows with r. The evaluation is now memory-bandwidth bound on the factor tables; the O(order) phase recomputation in the decay estimator is a smaller secondary term, left for later.

8 GPU sampling of the discretized Hamiltonian

The former non-goal N7 is lifted: the ab-initio engine runs on CUDA (CubeCL), and the port exists to *win on throughput*, unlike the earlier Fröhlich GPU port (a correctness vehicle). The physics favors the GPU here because the dominant cost is not pointer chasing but the vertex evaluation of Section 7: a contiguous rank-r complex dot streaming $\approx 16r$ bytes per call — a memory-*bandwidth* problem, and HBM bandwidth is the one resource where a GPU outclasses a CPU node by an order of magnitude. Measured on an A100 at production rank ($r = 800$, 50^3 grid), the cooperative evaluation sustains $\approx$ 1.24 TB/s ($\approx$ 80% of HBM peak; $\approx$ 90% at 20^3 where L2 helps), versus 14–76 GB/s for the same evaluation on CPU nodes.

Parallelism and the cooperative vertex dot. Chains never communicate, so the natural mapping is many independent chains per device. The microbenchmark that gated the port showed that *per-thread* evaluations never coalesce (0.4$\times$ a CPU node), so the kernel assigns **one chain per 32-lane warp**: every lane executes the chain's update logic *redundantly* — the RNG is counter-based (Philox4x32-10 keyed by (seed, chain, step)), so all lanes compute identical draws with no communication, control flow is warp-uniform by construction, and lanes cooperate only inside vertex evaluations, striding the rank axis with a shared-memory reduce. One arc factor $g(\mathbf{k}_e, \mathbf{q}, \nu)\, g^*(\mathbf{k}_a, \mathbf{q}, \nu)$ fuses both dots over a single shared $v_g(\mathbf{q}, \nu)$ strip, halving the dominant table traffic. Diagram state is a dense τ-sorted array per chain (positional keys mirroring the CPU engine one-to-one); one lane applies accepted mutations between barriers. Tables are stored planar (separate re/im f32), τ and all weight arithmetic are f64, and momenta are exact integer grid indices, as on the CPU.

Why lockstep execution is mandatory, and why it does not bias the sampling. The warp-uniformity above is a hardware *correctness* requirement, not a tuning choice. The cooperative reduce synchronizes the warp with block-wide barriers, and the hardware pairs barrier arrivals by *dynamic count*, not by source location: had lanes diverged, either some would wait at a barrier the others never reach (deadlock), or the counts would happen to match at *different* program points and lanes in unrelated phases would race through the one shared scratch buffer (silent bit-level corruption). Since the number of barriers a step executes depends on the update type, the proposal draws, and the diagram state, control flow must agree across lanes over the *entire dynamic trace* — which redundant execution of identical draws guarantees bit-for-bit. The statistical side is equally clean. The update-*type* schedule is drawn from a dedicated counter-based stream keyed by (seed, chain, step) — never by the state — and may even be *shared* by a group of chains (group-uniform selection, the variant used when several chains share a warp). Because the schedule is independent of the state and of every chain's private proposal/acceptance stream, each chain's *marginal* law is exactly that of the standard random-scan chain driven by the mixture kernel $n_{\mathrm{upd}}^{-1} \sum_u P_u$, and the $1/n_{\mathrm{upd}}$ selection probabilities of paired moves (phonon insertion $\leftrightarrow$ removal) cancel in the acceptance ratio exactly as in per-chain selection. Sharing the schedule alters only the *joint* law: chains attempting the same move class simultaneously are weakly correlated, which can only inflate the variance — a chain average is linear, so its expectation is untouched. The load-bearing invariant is that selection stays state-independent: an "impossible" proposal (e.g. a removal at order zero) counts as proposed-and-rejected and is never redrawn, since resampling until a possible move would make the effective selection distribution state-dependent and break the cancellation — a genuine bias, not a determinism nit.

Determinism and multi-GPU scaling. Because chain streams are keyed by *global* chain ids, a population split across ranks (one rank per GPU, chain offset = rank $\times\, C$) is a pure repacking of a single larger run: per-chain measurement batches concatenate in global chain order, and the merged report is **bit-identical** to the equivalent single-device run — verified on a 2×A100 node (2×8192 chains vs. one 16384-chain run, every estimator equal to the last bit). Multi-GPU throughput is therefore linear and the merged statistics are a pure function of (seed, total chains), independent of packing.

Validation stages. Correctness is certified in stages, each mechanical to check: (i) a flat, fixed-capacity CPU re-implementation of the four updates (the "executable spec") agrees with the production engine's acceptance ratios to 10^{-10} on imported random diagrams; (ii) the CUDA kernel, run at one lane per chain on CubeCL's CPU runtime, reproduces the flat driver *bit-for-bit* on the real LiF table (identical reports and accept counts); (iii) the same kernel passes exact accept-count parity on real hardware at 32 lanes; (iv) the physics anchor: the full 20^3 ladder configuration on one A100 gives $E_0 = -0.24826 \pm 0.00023$ eV against FEP-DMC's -0.2479 eV. All floating-point association in the kernel mirrors the CPU code term by term, so stage (ii) is exact rather than statistical.

Throughput at matched physics. At 20^3, rank 800, $\langle n \rangle \approx 16$, fixed $\tau = 60$ eV^{-1}: one A100 sustains 8.79×10^6 steps/s (kernel-only; a full production-statistics 20^3 ladder run is 151 s wall), versus 2.83×10^6 steps/s measured for the production CPU engine on a 128-core Zen3 node with the same configuration — 3.1× per GPU, 6.2× per two-GPU node. This is the CPU's *best* case: the 20^3 table (0.34 GB) fits the node's aggregate L3, while at 50^3 (5.4 GB) the CPU falls to DRAM bandwidth and the GPU keeps streaming from HBM, so the margin grows with the grid — exactly

where the D1 ladder needs it.

Estimator bookkeeping at GPU populations. Two scaling lessons from operating 10^4 chains. First, the leave-one-out jackknife must be computed as total − batch (O(bins) per replicate); recomputing the aggregate per replicate is $O(n_{\text{batches}}^2)$ and silently turns a two-minute run into hours at 10^5 batches. With chains independent, the natural batching is *one batch per chain*, jackknifing over chains. Second, warmup must exceed the diagram-order equilibration transient ($\sim 10^5$ steps for LiF at 20^3) *regardless* of how many chains amortize the statistics: a many-short-chain ensemble does not dilute the transient the way one long chain does. An early GPU run with 2×10^4 warmup steps per chain came out 1.7 meV shallow with $\langle n \rangle$ low by 2% — statistically resolved at 5σ against the long-chain CPU value — purely from under-equilibration; production GPU configs now warm up 3×10^5 steps per chain.

9 Practical pitfalls and fixes

Turning the EPW run into a *usable* vertex, and then a trustworthy DiagMC number, meant working through several failures whose symptoms were misleading. P1–P7 are data-preparation (A0/A1); P8–P9 are the DiagMC consumer (Phase B). They are recorded here so they are not rediscovered; the blow-by-blow A0/A1 diagnosis, with reproducing commands, lives in `datasets/LiF/EPH-NAN-DIAGNOSIS.md`.

P1 — Ultrasoft pseudopotentials give an all-`NaN` e–ph tensor (root cause). The first A0 deck used ultrasoft pseudopotentials (SSSP-efficiency GBRV). EPW's electron–phonon matrix elements are only supported with *norm-conserving* pseudopotentials — the ultrasoft augmentation charge is not carried into the coupling. The coupling came out finite *only* at $\mathbf{q} = \Gamma$ (the trivial reference q, which needs no symmetry rotation) and `NaN` at every other $\mathbf{q}$; the Bloch→Wannier sum then spread the `NaN` to 100% of `LiF.epmatwp`. Because `ph.x` *does* support ultrasoft, the phonons, dynamical matrix, $\mathbf{Z}_\kappa^*$ and $\boldsymbol{\varepsilon}_\infty$ were all correct — only the e–ph coupling was wrong, which is why the fault was easy to misattribute. *Fix:* switch Li and F to norm-conserving Pseudo Dojo ONCV (scalar-relativistic PBE). This raises the required plane-wave cutoff (`ecutwfc` : 50 → 96 Ry, `ecutrho` : 400 → 384 Ry, i.e. the norm-conserving 4×), which makes `ph.x` ∼3–4× slower (the wavefunctions move from the small smooth FFT grid onto the dense grid) — an unavoidable cost of the correct pseudopotential family. With norm-conserving pseudos the tensor is finite at every $\mathbf{q}$, and $\boldsymbol{\varepsilon}_\infty$, $\mathbf{Z}_\kappa^*$ are essentially unchanged.

P2 — "Right size, wrong values": a silent failure. The broken run finished cleanly and wrote `LiF.epmatwp` at exactly the expected byte size; the only hint was an easily-dismissed `IEEE_INVALID` note at exit. The file was 100% uninitialised-memory `NaN` (a uniform garbage word that even *changed* between reruns). An earlier "A0 done" had been declared on file size alone, without ever checking the tensor *values*. *Fix:* the pipeline (`run_pipeline.nu`) now runs an explicit finiteness check on `LiF.epmatwp` and aborts loudly on `NaN`, so a broken run can no longer masquerade as a good one.

P3 — Two dead ends from assuming an upstream bug. Before the pseudopotential cause was found, two "EPW bug" hypotheses were pursued and *both failed*: (i) widening the Wannier outer window `dis_win_max` to enclose every band made the coarse tensor *worse*, not better; (ii) a bounds-guard patch to EPW's `gmap_sym` routine (a genuine latent out-of-bounds read) compiled in cleanly yet left the output byte-identical — so that read was not the cause. The lesson, recorded explicitly: suspect the *inputs* (here, an unsupported pseudopotential family) before patching upstream code.

P4 — The reference `.epb` reader has the wrong record layout. The Python reference (`patt126`) reads the coarse Bloch `.epb` file assuming a single band count throughout the record. The real record interleaves a *20-band* eigenvalue array with a *15-band* coupling array (of which only a 5×5 block is populated); a naïve parse silently mis-aligns and yields a *false positive* — a "the coarse data looks fine" reading that briefly sent the debugging down a wrong path. *Fix:* the true record layout was reverse-engineered and byte-verified — a *single* unformatted Fortran record (4-byte length markers) [n_{qc}, $\mathbf{x}_{qc}(3, n_{qc})$, $\varepsilon(20, n_k)$, $D(6, 6, n_{qc})$, $g(15, 15, n_k, 6, n_{qc})$, $\mathbf{Z}^*, \boldsymbol{\varepsilon}_\infty$], all little-endian and column-major, one file per pool with the pool-local n_k recovered from the record length. Two indexing traps: the coupling's mode axis sits *interior* (stride $n_k \cdot 6$, not $6 \cdot n_k$), and the conduction-band minimum is kept-band local index 0 in the coupling array but eigenvalue index 5 in ε. The Rust reader uses this directly and decodes only the needed conduction-band slice on demand.

P5 — epw coarse-grid prerequisites (reproducibility). Several setup constraints are load-bearing and non-obvious: the `nscf` k-points must be an *explicit* Γ-centred crystal list (an automatic, half-shifted Monkhorst–Pack mesh breaks EPW's $\mathbf{k}+\mathbf{q}$ folding); the whole `scf/ph/nscf/epw` chain must use *one* Quantum ESPRESSO version (a mixed-version `phsave` corrupts the heap); and EPW needs a large stack (`ulimit -s unlimited`). Finally, changing the pseudopotentials shifts the band edges (∼0.15 eV here), so the disentanglement windows (`dis_froz_max`, `dis_win_max`) must be re-tuned against the new `nscf` eigenvalues — otherwise the frozen conduction-band minimum, which must stay exact, can slip out of its window.

P6 — The polar term lives in the Cartesian displacement basis, and carries an electron-band rotat
Two subtleties in Eq. (3) decide whether the split $g = g^{\mathcal{S}} + g^{\mathcal{L}}$ of Eq. (2) actually closes, and both were found by round-tripping against the coarse `.epb` (which stores the *full*, pre-subtraction coupling). *(i) Basis.* EPW subtracts $g^{\mathcal{L}}$ from the coupling in the *Cartesian atomic-displacement* basis ($\kappa\alpha$, i.e. per atom and Cartesian direction), *before* the phonon-eigenvector projection $\mathbf{e}_{\kappa\nu}(\mathbf{q})$ and the $(\hbar/2M_\kappa\omega)^{1/2}$ mode factor. So the reconstructed short-range part and the re-added $g^{\mathcal{L}}$ must both be formed in that basis and summed there; projecting onto branches first is inconsistent. The mode factor and eigenvector projection are applied afterwards by the consumer. *(ii) Gauge.* The band-rotation factor $[U_{\mathbf{k}+\mathbf{q}} U_{\mathbf{k}}^\dagger]_{mn}$ — at the Γ-slice the conduction-band Bloch overlap $\langle u_{\Gamma+\mathbf{q}} | u_\Gamma \rangle$ — is easy to drop under the "vertex depends on $\mathbf{q}$ only" reduction (both electron legs rotated with U_Γ, overlap $\equiv 1$). Dropping it puts the polar term in the wrong electron gauge: near Γ the reduced sum still overshoots $\|g\|$ by $\sim 9\%$, and the overlap itself falls from ≈ 0.94 at the first coarse shell to 0 at the zone boundary as the conduction-band character changes with $\mathbf{k}$. Restoring the exact bra ($u_{\Gamma+\mathbf{q}}$) and the overlap makes the decomposition exact. *Result:* against all 215 non-zero coarse $\mathbf{q}$, $g_{\text{full}} = g^{\mathcal{S}} + \langle u_{\Gamma+\mathbf{q}} | u_\Gamma \rangle \, g^{\mathcal{L}}$ to machine precision — gauge-invariant cosine = 1.000000 and $|\text{ratio} - 1| \sim 3 \times 10^{-7}$ (the `.epb` Kohn–Sham states carry an arbitrary per-$\mathbf{k}$ phase, so the check is on direction and magnitude, not the raw complex difference). The reduced $\mathbf{q}$-only vertex remains the object the sampler consumes; the exact bra + overlap is used only to prove the reconstruction and the polar term against real data.

P7 — The dynamical matrix carries the same polar long-range. The subtraction of P6 is not confined to the e-ph vertex: EPW removes the rigid-ion dipole–dipole term from the dynamical matrix as well (`rgd_blk`), so the Wannier-interpolated phonons are short-ranged — the optical triplet stays degenerate at Γ at an unphysical 11.95 THz with no LO–TO splitting, where the physical TO is 8.12 THz (agreeing with `ph.x`). This is easy to overlook because the phonon *eigenvectors* are barely affected (the optical modes are symmetry-fixed

Li–F relative motion), but the *frequency* enters the vertex through the zero-point factor $(2M_\kappa\omega_{\mathbf{q}\nu})^{-1/2}$, so a short-range ω mis-scales $|g|^2$ by a factor of order two. *Fix:* port EPW's `rgd_blk` (Gonze form, Eq. (4)) – the $\mathbf{q} \neq 0$ dipole–dipole sum over all atom-pair blocks plus the $\mathbf{q}$=0 self-term on the diagonal blocks – and add it back before diagonalizing. *Validated* at two levels: the reconstructed long-range term equals coarse-`dynq` minus the short-range force constants *element-by-element* (ratio 1.000), and all six branches then reproduce the DFPT `dynq` at the coarse q ($[2.76, 2.76, 5.23, 8.12, 8.12, 18.45]$ THz, LO = 18.45) — the phonon analog of the P6 round-trip. The restored $\omega_{\mathrm{LO}}(\Gamma) = 18.67$ THz gives a Lyddane–Sachs–Teller ratio $\varepsilon_0/\varepsilon_\infty = (\omega_{\mathrm{LO}}/\omega_{\mathrm{TO}})^2 \approx 5.3$, i.e. $\varepsilon_0 \approx 11$ (experiment ≈ 9).

P8 — The phonon quantization factor was dropped in the branch projection (Phase B). Distinct from P7 (where the *frequency* was wrong): here $\omega_{\mathbf{q}\nu}$ was correct, but the consumer projected the Cartesian coupling onto phonon branches with only the eigenvector $\mathbf{e}_{\kappa\nu}$ and the Ry→eV conversion, *omitting* the $(2M_\kappa\omega_{\mathbf{q}\nu})^{-1/2}$ zero-point factor of Eq. (3) entirely. Every $|g|$ was then $\sim 7\times$ too large, so the second-order energy came out ≈ -11 eV ($\sim 30\times$ too deep) and the DiagMC diagram order ran away (no order-0 samples, the estimator's normalization sector). The bug hid because the *oracle* used for the SVD cross-check made the same omission, and the `.epb` test validates only the *Cartesian* coupling, above the projection. *Fix:* apply $(2M_\kappa\omega_{\mathbf{q}\nu})^{-1/2}$ in the branch projection of *both* the tabulated vertex and the direct oracle, guarding the $\omega \to 0$ acoustic modes at Γ. *Validated:* $E^{(2)}$ fell to -0.10 eV on the 8^3 grid — the expected $\sim 3.5\times$ coarse-grid underestimate of the continuum $-\alpha\hbar\omega_{\mathrm{LO}} \approx -0.37$ eV — and the sampler then equilibrates at order ≈ 3.

P9 — Reading E_0 from the decay estimator: μ and $\tau_{\max}$ (Phase B). The ground state is the large-τ slope of $G(\tau) \sim Z\, e^{-(E_0-\mu)\tau}$, and two knobs must bracket it. The chemical potential μ must sit *just below* E_0: if $\mu > E_0$ the weight grows with τ and the diagram order runs away (symptom: "no order-0 samples"); if μ is far below E_0 the phonon cloud is starved and the fit window falls where G is exponentially small, so E_0 scatters with underestimated error bars. The window must also satisfy $\tau_{\mathrm{fit}} \gg 1/(E_1 - E_0) \sim 1/\omega_{\mathrm{LO}}$ to clear excited-state contamination. The acceptance criterion is that the extracted $E_0 = \mu + \text{slope}$ is μ-*independent*. On the coarse 8^3 grid the polaron is effectively strong-coupling (order 3–5, $E_0 \approx -0.26$ eV), so $\mu \approx -0.5$ eV with $\tau_{\max} \approx 30$ eV^{-1} is the working regime.

10 The finite-β thermodynamic energy estimator (FEP-DMC)

The energy in Section 9 (P9) is read from the large-τ decay of $G(\tau)$, which carries a documented sensitivity to the fit window $\tau_{\max}$ at strong coupling. FEP-DMC (Luo, Park, Bernardi, *Nat. Phys.* **21**, 1275 (2025); code a Perturbo fork) instead measures the polaron energy as a *thermodynamic* (operator) expectation value on each worldline, with *no* fit. This section records how that estimator works and its single-band form. It is **implemented** as the production estimator: the sampler runs in a fixed-τ ensemble (external time pinned at $\tau_{\max}$), where the chemical potential μ drops out of the ensemble entirely — eliminating the μ-bracketing of P9 — and the local energy $E_{\mathrm{loc}} = \tau^{-1}\left[\sum_{\mathrm{seg}} \varepsilon\, \Delta\tau + \sum_{\mathrm{arcs}} \omega\, \Delta\tau - 2N_{\mathrm{arcs}}\right]$ is averaged with the sign. It is validated against an exactly solvable dispersionless (Lang–Firsov) lattice model, $E_0 = -\bar{g}^2/\omega$ at all couplings, and against the Fröhlich $\alpha = 1, 2$ benchmarks, where it agrees with the decay fit at lower variance.

Free-τ vs. fixed-τ ensembles. Both estimators sample diagrams of the zero-temperature imaginary-time Green's function

$$G(\mathbf{p},\tau) \;=\; \langle 0|\, c_{\mathbf{p}}\, e^{-\tau(\hat{H}-\mu)}\, c_{\mathbf{p}}^{\dagger}\,|0\rangle \;\xrightarrow{\tau\to\infty}\; Z\, e^{-(E_0-\mu)\tau}, \tag{5}$$

a diagram being one electron backbone of external length τ dressed by phonon arcs with internal vertex times and momenta. The two modes differ in whether that *external* τ is itself a sampled variable. In the *free-τ* ensemble (`fixed_tau: false`) the Markov chain updates $\tau \in (0, \tau_{\max})$ alongside the internal degrees of freedom; histogramming the signed weights over τ (normalized through the analytically integrable order-zero sector) yields the whole curve $G(\tau)$ in one run, from which the decay fit extracts both E_0 and the quasiparticle weight Z, with the per-bin local-energy curve $E(\tau)$ as a built-in convergence check. Its costs are those of Section 9, P9: statistics diluted over the τ bins with only the large-τ tail informative, a genuine fit-window systematic, and a load-bearing μ — the weight carries $e^{\mu\tau}$, which shapes the sampled τ distribution. In the *fixed-τ* ensemble (`fixed_tau: true`, FEP-DMC's `zeroTMC`; all production runs) the external time is pinned at $\tau_{\max}$ and only internal degrees of freedom are sampled: $e^{\mu\tau}$ is a constant that cancels in every ratio, so μ drops out entirely, and all statistics sit at the single τ that matters. The energy then follows from the internal-time rescaling identity: a diagram of N arcs has $2N$ internal vertex times, so rescaling them by τ exposes the explicit τ-dependence $\tau^{2N} e^{-\sum \varepsilon\,\Delta\tau - \sum \omega\,\Delta\tau}$ of each configuration's weight, whence

$$-\partial_\tau \ln G(\tau) \;=\; \frac{1}{\tau}\Big\langle \sum_{\text{seg}} \varepsilon_i\,\Delta\tau_i \;+\; \sum_{\text{arcs}} \omega_a\,\Delta\tau_a \;-\; 2N \Big\rangle_{\tau} \;=\; E_0 - \mu + \mathcal{O}\big(e^{-\Delta\tau}\big), \tag{6}$$

the sign-weighted average taken in the fixed-τ ensemble. This is the E_{loc} quoted above (the code's `local_energy`), and equals Eq. (9) in the $T \to 0$ single-band limit up to the vertex-counting convention ($-2N$ arcs here vs. $-(N_v - 1)$ vertices there). The estimator therefore approaches E_0 *from above* at finite τ, as $\sum_i w_i \Delta_i e^{-\Delta_i \tau}$ over excited states i of the $\mathbf{p}$ sector with gaps Δ_i and relative weights w_i.

The relevant excited states at $\mathbf{p} = \Gamma$ are the polaron at $-\mathbf{q}$ plus one phonon at $\mathbf{q}$, with gap $\Delta(\mathbf{q}) = E_0(-\mathbf{q}) - E_0 + \omega_\nu(\mathbf{q})$. For the LO branch, $\Delta \geq \omega_{\text{LO}} = 77.2$ meV (18.67 THz), which gives $\omega_{\text{LO}}\tau \approx 4.6$ at $\tau = 60$ eV^{-1} and ≈ 18.5 at $\tau = 240$ eV^{-1} (FEP-DMC's effective $1/k_BT$ at 50 K is 232 eV^{-1}). The acoustic branches are gapless, however: on the n^3 grid the smallest gap is $\approx c_s q_{\min} \propto 1/n$ (4.4 meV at 16^3, below 1 meV beyond $n \approx 70$, where the soft-mode cutoff removes those modes). Convergence in τ is therefore *not* uniform in n. The weak acoustic coupling keeps the amplitude small, but no single $\Delta \approx \omega_{\text{LO}}$ controls it. Measured: $E_0(\tau 240) - E_0(\tau 60) = -0.9 \pm 0.3$ meV at 20^3 but -2.8 ± 0.1 meV (flat) for $n \geq 40$. Because fixed-τ produces no curve whose flattening can be inspected, two further checks certify τ convergence: a $\tau = 480$ eV^{-1} guard at $n = 120$, and a free-τ run at 40^3. The free-τ run bins the same E_{loc} over $\tau \in (0, 240]$; its per-bin average must reproduce the fixed-τ points at $\tau = 60, 240$, and its shape measures the effective gap directly.

Quantity measured. Rather than the asymptotic slope, the polaron energy is the operator average

$$E_{\text{pol}} \;=\; \langle \hat{H} \rangle \;=\; \frac{\mathrm{Tr}\big[\hat{H}\, e^{-\beta\hat{H}}\big]}{\mathrm{Tr}\big[e^{-\beta\hat{H}}\big]}, \tag{7}$$

evaluated on each Monte-Carlo worldline of total imaginary-time length $\tau = \beta$, where $\beta = 1/k_BT$ is the code's `tauMax`. Here Tr is the matrix trace $\mathrm{Tr}[M] = \sum_i M_{ii}$; the trace over the complete

basis is what turns Eq. (7) into a proper thermal (and, as $\beta \to \infty$, ground-state) expectation value. This is the standard Prokof'ev–Svistunov / Mishchenko thermodynamic estimator.

Worldline weight. A configuration is one electron backbone on $\tau \in [0, \beta]$, cut into segments by vertices at $\tau_1 < \tau_2 < \cdots$, with phonon arcs joining vertex pairs. In single-band form its weight is a product of

- electron segments s of duration $\Delta\tau_s$: $e^{-\varepsilon_{k_s}\Delta\tau_s}$ (the code's `Gel`$(\varepsilon,\tau) = e^{-\varepsilon\tau}$ and `propagator`). The electron is kept at $T = 0$; only phonons are thermal, as appropriate for a single carrier;
- phonon arcs a of span $\Delta\tau_a$ at frequency ω_a: the finite-T propagator $D(\omega,\tau)$ below;
- a vertex factor $g_{mn\nu}(\mathbf{k},\mathbf{q})$ at each of the N_v vertices.

Thus $\ln w$ depends on β through every segment length, every arc span, and N_v.

The estimator as a time average. With $\hat{H} = \hat{H}_{\rm el} + \hat{H}_{\rm ph} + \hat{H}_{\rm int}$, the key move is *not* to read the energy only at the endpoint $\tau = \beta$ but to use the ground-state-dominated identity

$$\langle \hat{H} \rangle \;=\; \frac{1}{\beta}\int_0^\beta \langle \hat{H}(\tau)\rangle \, d\tau, \tag{8}$$

i.e. the *local* energy is averaged over the entire worldline. This is the variance reduction: β times more information per configuration than a single asymptotic slope. Each term of $\hat{H}$ maps to one accumulator, and the three sum to

$$E_{\rm pol} \;=\; \frac{1}{\beta\,\langle s\rangle}\Big\langle \underbrace{\sum_{\rm seg}(\varepsilon_i+\mu)\,\Delta\tau_i}_{\langle H_{\rm el}\rangle} \;+\; \underbrace{\sum_{\rm arcs}\tfrac{1}{2}\,\omega_a|\Delta\tau_a|}_{\langle H_{\rm ph}\rangle} \;-\; \underbrace{(N_v-1)}_{\langle H_{\rm int}\rangle}\Big\rangle, \tag{9}$$

with $\langle s\rangle$ the average sign. The three pieces are:

Band energy $\langle H_{\rm el}\rangle$. The code inserts the operator via `Hpropagator`, whose diagonal is $e^{-\varepsilon\tau}(\varepsilon+\mu)\tau$. The extra factor $\varepsilon\,\Delta\tau$ is exactly $\int_0^{\Delta\tau} ds\,\varepsilon\,e^{-\varepsilon\Delta\tau} = \varepsilon\,\Delta\tau\,e^{-\varepsilon\Delta\tau}$ — the "$\hat{H}e^{-\hat{H}\tau}$" insertion integrated over the segment. Summed over segments and divided by the diagram weight it yields $\frac{1}{\beta}\int_0^\beta \varepsilon_{k(\tau)}\,d\tau$. Energies are stored band-bottom subtracted (ε) with $\mu = \min\varepsilon$, so $\varepsilon+\mu$ restores the true band energy.

Phonon energy $\langle H_{\rm ph}\rangle$. Each arc present over $\Delta\tau_a$ contributes ω_a to the instantaneous phonon energy; time-averaging gives $\frac{1}{\beta}\sum_a \omega_a\Delta\tau_a$. The $\frac{1}{2}$ corrects double counting of an internal arc seen from both its vertices; external (head/tail-anchored) phonon lines carry no $\frac{1}{2}$.

Interaction energy $\langle H_{\rm int}\rangle$. In the bare imaginary-time expansion the coupling-energy estimator evaluates to $-\langle N_v\rangle/\beta$ (each vertex contributes $-1/\beta$); the "-1" offset is the head-vertex normalization convention.

Equation (9) is thus just $\langle H_{\rm el}\rangle + \langle H_{\rm ph}\rangle + \langle H_{\rm int}\rangle = \langle\hat{H}\rangle$, measured at $\tau = \beta$ with no fitting window.

Finite-β phonon propagator. The arcs use the thermal bosonic propagator

$$D(\omega,\tau) \;=\; \frac{e^{-\omega\tau}+e^{-\omega(\beta-\tau)}}{1-e^{-\omega\beta}}, \tag{10}$$

whose first term $e^{-\omega\tau}$ is spontaneous emission (surviving at $T=0$), second term $e^{-\omega(\beta-\tau)}$ is thermal absorption / wrap-around, and denominator $1/(1-e^{-\omega\beta}) = 1+n_B(\omega)$ is the Bose enhancement. As $\beta\to\infty$ it collapses to $e^{-\omega\tau}$, which is what the code's `zeroTMC` flag hard-codes — and the LiF example uses `zeroTMC`, so LiF(e) is effectively the $T=0$ limit even though the machinery is finite-T. Soft modes below a `phfreq_cutoff` ≈ 1 meV are dropped ($D\to 0$) to avoid the $1/(1-e^{-\omega\beta})$ and $(2M_\kappa\omega)^{-1/2}$ divergences.

Matrix (multiband) form. In multiband mode the segment propagators, vertices, and the $\hat{H}$-insertion are all matrices, so the diagram weight is $\mathrm{Tr}\big[\prod_i g_i\, e^{-H\Delta\tau_i}\big]$ and inserting $\hat{H}$ at one segment needs the product of everything to its left and right (the `left_env`/`right_env` environment matrices), with per-step rescaling. The sign is carried as $\mathrm{sign}(\mathrm{Re\,Tr})$ and divided out via $\langle s\rangle$. In single band all of this collapses to the scalars of Eq. (9); the trace is trivial.

Why it replaced the decay fit. Relative to the decay fit of P9, Eq. (9) removes the $\tau_{\max}$ window arbitrariness (the whole $[0,\beta]$ worldline is used), lowers variance (it time-averages rather than reading one slope), and has a systematic error controlled by β (excited-state contamination $\sim e^{-\Delta\beta}$). The single-band form accumulates the three sums of Eq. (9) per sample, dividing by β and $\langle s\rangle$ — no environment matrices are needed until the multiband (hole/degenerate) work of D2. It has been cross-validated against the Fröhlich $\alpha = 1, 2$ Mishchenko values (stage 0), where it agrees with the decay estimator, and against the exactly solvable dispersionless lattice model noted above; all production numbers (Section 6) use it.

11 Phonon-cloud estimators from central time slices

The energy estimator is exact at every τ because it is a log-derivative, $E_{\mathrm{loc}} = -\partial_\tau \ln G$. Occupation-type observables have no such identity. Inserting an operator $\hat{O}$ at time t on the worldline gives

$$O(t) = \frac{\langle 0|c_{\mathbf{p}}\, e^{-\hat{H}(\tau-t)}\,\hat{O}\, e^{-\hat{H}t}\, c_{\mathbf{p}}^{\dagger}|0\rangle}{G(\mathbf{p},\tau)} \;\xrightarrow{\;t,\,\tau-t\gg 1/\Delta\;}\; \langle\Psi_0(\mathbf{p})|\hat{O}|\Psi_0(\mathbf{p})\rangle, \tag{11}$$

and the ground-state value is reached only far from both ends. In the bare-basis expansion the intermediate state at t is the electron segment crossing t plus the arcs spanning t. Hence, for $\hat{O} = \hat{P}_N$ (projector on N phonons), $O(t)$ is the sign-weighted probability that exactly N arcs cross t, i.e. FEP-DMC's $Z_{N,\mathbf{p}}$; $\hat{N}_{\mathrm{ph}}$ and its per-branch parts count the crossing arcs, and $\hat{n}_{\mathbf{k}}$ reads the crossing segment's momentum ($f_e(\mathbf{k})$ of their Eq. (4)). None of these needs external phonon lines or a new update.

The averaging window must be central. The whole-worldline average $\tau^{-1}\sum_a \Delta\tau_a$ is biased low by the ends, where the cloud is still building from $c^\dagger|0\rangle$. In the dispersionless model $N(t) = \bar{N}(1-e^{-\omega t})(1-e^{-\omega(\tau-t)})$, so the full average is $\bar{N}[1-2(1-e^{-\omega\tau})/(\omega\tau)+e^{-\omega\tau}]$: an $O(1/\omega\tau)$ error, about -11% for LiF at $\tau = 240$ eV^{-1}. The production estimator averages Eq. (11) over $t\in[\tau/2-w,\tau/2+w]$ (default $w=\tau/8$), which converges as $e^{-\Delta(\tau/2-w)}$ and is certified by w-independence. Implementation plan: `plans/open/nph-estimator.md`.

12 Polaron dispersion, effective mass, and the acoustic-emission threshold

Protocol. $E_0(\mathbf{p})$ is measured on grid momenta along $\Gamma \to X$ ($\mathbf{p} = \frac{j}{n}(\mathbf{b}_1 + \mathbf{b}_3)$) and $\Gamma \to L$, each point a separate fixed-τ run (`ironmc dispersion sweep`; runbook `notebooks/LiF-dispersion-mass.ipynb`). Cubic symmetry forces $E_0(\mathbf{p}) = E_0 + |\mathbf{p}|^2/2m_p + c_{4,\mathrm{leg}}|\mathbf{p}|^4 + \dots$ with one scalar m_p (`latex/symmetry-paths`). Fit windows are fixed in physical $|\mathbf{p}|$ (Å^{-1}), never in grid index: at fixed j the momentum shrinks as $1/n$, and an index window mistakes non-parabolicity for grid dependence. The mass is reported as m_p/m_b, where the band mass $m_b = 0.6926\, m_e$ is the Γ curvature of the same Wannier $\varepsilon(\mathbf{k})$ the sampler uses (`examples/band_mass.rs`). FEP-DMC's "m_p/m_e" uses exactly this normalization; m_e there is the band effective mass.

Current result. With the $p^2 + p^4$ model over $|\mathbf{p}| \le 0.16$ Å^{-1}: $m_p/m_b = 5.31 \pm 0.23\,(\mathrm{stat}) \pm 0.42\,(\mathrm{window})$ at 80^3 and $5.0 \pm 0.3 \pm 1.0$ at 120^3. At the same $|\mathbf{p}|$, 40^3, 80^3 and 120^3 agree to within about 1 meV, so the dispersion converges in n much faster than E_0. A pure parabola is excluded beyond $|\mathbf{p}| \approx 0.1$ Å^{-1} (χ^2 grows, the fitted mass drifts upward): parabolic fits give 6.0–6.4 over $|\mathbf{p}| \le 0.12$ Å^{-1} and up to ≈ 10 over wider ranges. FEP-DMC's 6.1 (Γ–L, parabolic fit) is therefore consistent with our dispersion, but a single parabolic number is window-dependent at the ± 1 level.

The acoustic-emission (Landau) threshold. For each total momentum $\mathbf{p}$, the fixed-τ estimator converges as $\tau \to \infty$ to the *lowest* energy in that sector:

$$E_{\mathrm{sector}}(\mathbf{p}) = \min\Big\{ E_{\mathrm{qp}}(\mathbf{p}),\ \min_{\mathbf{q},\nu}\big[E_{\mathrm{qp}}(\mathbf{p}-\mathbf{q}) + \omega_\nu(\mathbf{q})\big],\ \dots \Big\}. \tag{12}$$

The second entry is the polaron recoiling to $\mathbf{p} - \mathbf{q}$ after emitting one real phonon. For LO phonons it is the familiar flattening at $E_0 + \omega_{\mathrm{LO}}$. For acoustic phonons, $\omega \approx c_s q$ with no gap. With $E_{\mathrm{qp}} \approx E_0 + p^2/2m_p$, the emission state lies below the quasiparticle as soon as the polaron's group velocity exceeds the sound velocity, $|\mathbf{p}|/m_p > c_s$: the lattice analogue of Cherenkov radiation. Beyond that momentum the bottom of the sector is $\approx E_0 + c_s|\mathbf{p}| - m_p c_s^2/2$. For LiF, $\omega_{\mathrm{TA}} = 4.93$ meV at $|\mathbf{q}| = 0.155$ Å^{-1} along $\Gamma \to X$ gives $c_s \approx 32$ meV Å (4.8 km/s). With $m_p \approx 3.7\, m_e$ this puts the threshold at $p_c = m_p c_s \approx 0.015$ Å^{-1}, below the first nonzero $\Gamma \to X$ grid point even at 120^3 (0.026 Å^{-1}).

The quasiparticle becomes a resonance that can decay by acoustic emission. The measured $E_0(\mathbf{p}, \tau)$ remains quasiparticle-dominated only while the admixture of the lower continuum, $\sum_i w_i e^{+\delta_i \tau}$ with $\delta_i = E_{\mathrm{qp}} - E_i > 0$, stays small. Each weight is small, $w_i \sim |g_{\mathrm{ac}}|^2/(N_q \delta_i^2)$ (acoustic coupling is weak and each state carries $1/N_q$), but the factor $e^{\delta_i \tau}$ grows with τ and with $|\mathbf{p}|$. So, unlike at Γ, longer τ is not automatically better at finite $\mathbf{p}$. The diagnostic is cheap: at the same $(n, \mathbf{p})$, $E_0(\mathbf{p}, \tau 240) - E_0(\mathbf{p}, \tau 60)$ must equal the Γ shift (-2.8 meV) if the quasiparticle dominates, while an excess that grows with $|\mathbf{p}|$ signals contamination. Keeping mass fits at small $|\mathbf{p}|$, where δ is a few meV, limits the effect in any case.

13 Remote bands: the Allen–Heine–Cardona correction

The DiagMC is exact for the Hamiltonian $\hat{H}_n$ it samples: one Wannier band (the CBM), linear e–ph coupling. Two pieces of the physical band-edge renormalization are outside it: (i) Fan–Migdal virtual transitions into bands not included in the sampled manifold, and (ii) the Debye–Waller term (second order in the displacement at one vertex), which a linear-coupling DiagMC never

generates. FEP-DMC adds both perturbatively (AHC, their ref. 36, 37), moving LiF(e) from -0.408 to -0.538 eV. This is a 130 meV correction, larger than every DiagMC systematic above. Consistently with the sampled manifold,

$$\Delta E_{\mathrm{AHC}} = \mathrm{Re}\,\Sigma^{\mathrm{FM}}_{\mathrm{CBM}}\big(\varepsilon_{\mathrm{CBM}}\big)\Big|_{m\notin\{\mathrm{CBM}\}} + \Sigma^{\mathrm{DW}}_{\mathrm{CBM}}, \qquad \Sigma^{\mathrm{FM}}_{n\mathbf{k}}(\varepsilon) = \frac{1}{N_q}\sum_{m\nu\mathbf{q}} |g_{mn\nu}(\mathbf{k},\mathbf{q})|^2 \,\frac{1}{\varepsilon - \varepsilon_{m\mathbf{k}+\mathbf{q}} - \omega_{\nu\mathbf{q}}}, \tag{13}$$

at $T = 0$ (single carrier). The band sum runs over all Kohn–Sham bands, via Sternheimer for the high ones, and excludes the intraband CBM→CBM term that the DiagMC already contains to all orders. The practical route and its pitfalls are in `plans/open/ahc-correction.md`.